

Ayush Bhagwat

Backend Engineer (Go) · AI agents · trading systems

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Summary

Backend engineer (Go). At Radisys I built the service that provisions bare-metal servers into Kubernetes clusters for Deutsche Telekom's ~900-site broadband rollout. Now building backends where correctness is the hard part: an AI support agent whose backend, not the model, enforces the rules, and a perpetual-futures exchange. Seeking backend and backend-AI roles (SDE-1/SDE-2), remote or India.

Experience

Software Engineer, Backend

Radisys | Deutsche Telekom Access 4.0, cloud-native FTTH broadband platform

Jan 2023 – Feb 2024

Bengaluru, India

- Built the **Pod Bring-up Service (PBS)**, a stateless **Go/Gin microservice** that provisions bare-metal servers into Rancher-managed **K3s clusters** for Deutsche Telekom's ~900-site FTTH rollout: DNS/DHCP setup, IPMI PXE boot, SUSE OS install via Cobbler/AutoYaST, K3s install via Salt, and Rancher registration.
- Made every bring-up step **safe to retry**: workflow state lives in **MongoDB** and each step deletes before it creates, so a crashed or re-triggered install resumes without duplicate DNS records, OS profiles or inventory; installing nodes report progress back to PBS through callbacks.
- Built the **site decommissioning flow**, tearing a site down in dependency order: Rancher cluster, SUSE Manager systems and Salt keys, install profiles, DNS/DHCP records, BMC power-off.
- Built **stub services** mimicking vendor APIs (Rancher, SUSE Manager, IPMI/BMC), so the control plane runs in **GitLab CI** without hardware; documented every PBS workflow in 5 sequence diagrams.
- Deployed and ran **Terraform**-provisioned virtual PODs (the full stack on VMs), so squads without physical hardware could test end to end.

Go, Gin, MongoDB, Kubernetes (K3s), Rancher, Helm, SUSE Manager, IPMI, GitLab CI, Terraform

Projects

AI Customer Support Agent

github.com/MidNight91119/ai-support-agent

Go · LLM tool calling · concurrency

- An AI agent works each ticket by calling backend tools (order lookup, policy search, stock check), then resolves it, asks for details, waits for approval, or escalates to a human.
- The backend enforces the rules, not the model**: customers touch only their own orders, refunds above a limit need human approval, each order gets at most one refund or replacement, and every argument the model writes is validated before it runs.
- Safe under concurrent requests**: state changes happen in one short locked section that never waits on the model, so no order is refunded twice; if the AI provider is down, the ticket fails safely with no side effects.

Perpetual Futures Exchange

github.com/MidNight91119/perps-exchange

Go · order book · Binance feed

- Matching engine with price-time priority**: limit and market orders, best price then oldest first, partial fills, post-only.
- Margin and leverage**: locks collateral before trading, tracks positions and PnL, and **liquidates on mark price**, not last trade.
- Funding payments** that tie the contract's price to spot, using a live index price from Binance.

SimpleBank

github.com/MidNight91119/simplebank

Go · gRPC · PostgreSQL · Redis · AWS EKS

- Payments backend with **deadlock-safe concurrent transfers**: one database transaction per transfer, rows locked in a fixed order, every money movement recorded in a ledger.
- Same API over **gRPC and REST**, PASETO/JWT auth with roles, a Redis-queued email worker; CI against real Postgres, and every push to main deploys a Docker image to **AWS EKS**.

Skills

Languages: Go, SQL, TypeScript, Python

Backend: REST, gRPC, Gin, net/http, PostgreSQL, sqlc, MongoDB, Redis

AI: LLM tool calling, agent loops, OpenAI-compatible APIs

Infrastructure: Docker, Kubernetes, Helm, Terraform, AWS (EKS, ECR), GitHub Actions, GitLab CI

Education

B.Tech, Information Technology | CGPA 8.27

Indian Institute of Information Technology, Lucknow (IIIT Lucknow)

2019 – 2023

Lucknow, India

Achievements

- 1000+ problems solved** on Codeforces, AtCoder and LeetCode | **1st place**, Hack-O-Heist hackathon, IIIT Lucknow | **Best Presentation Award**, Radisys 2023.
- VP Membership, Toastmasters Radisys (2023) | E-Cell Coordinator, IIIT Lucknow (2022) | Hult Prize Campus Director (2021).